MedNear: Project Overview

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Executive summary

MedNear is an AI-powered platform that shows patients which nearby pharmacies have their medicine in stock, and lets pharmacies keep inventory current by scanning bills instead of typing.

Patients search by medicine name or upload a prescription, then see pharmacies on a map with availability, stock level, price and distance. They can hold the medicine before travelling. Pharmacy owners scan purchase invoices with OCR, review the extracted items, and confirm the stock update. They also set their own selling prices and margins within regulations.

What makes MedNear different is the closed loop between the two sides: pharmacy scans feed what patients see, and what patients fail to find feeds back to pharmacies as demand signals.

Problem statement

Patients lose time and sometimes treatment days visiting several pharmacies to find one prescribed medicine, and pharmacies lose sales to stock-outs and money to expired stock.

For patients

- No reliable way to know which pharmacy has a medicine before travelling.
- Prices vary between pharmacies and are hard to compare.
- A prescription with several medicines may need visits to more than one shop.
- When a brand is missing, there is no safe, easy way to find a suitable substitute.

For pharmacies

- Many independent pharmacies still track stock on paper or spreadsheets.
- Typing every purchase bill into a system is slow and error-prone.
- Owners cannot see what local customers are looking for and cannot find.

Existing systems and gaps

Existing tools solve either the patient search or the pharmacy inventory, not both together. The table is based on public descriptions of these products and was not exhaustive, so verify it before formal use.

Existing system	What it does	Side served
NearByMedi	Finds the nearest store stocking all listed medicines; West Bengal, scraped database	Patient
MedLoc	Location-based finder with alternatives by chemical composition	Patient
MedPlus Mart	Real-time availability across MedPlus stores	Patient (single chain)
GdziePoLek	Availability, price and reservation for pickup (Poland)	Patient
Janaushadhi Sugam	Generic vs branded price comparison	Patient
Pharmixa	AI bill scanner, batch and expiry tracking	Pharmacy
Minfos AI Invoice Scanner	Invoice scanning with review before ordering	Pharmacy
Aushadhi Vault	Prescription scanning for billing, inventory agent	Pharmacy

The seven gaps MedNear fills:

#	Gap in existing systems	MedNear feature
1	Patient finders and pharmacy tools are separate	One platform: pharmacy bill scans update the stock patients search
2	"Available" has no reliability measure	Stock confidence score, last-verified time, confirm ping
3	Real-time stock is limited to a single chain	Works across independent pharmacies via OCR updates
4	Alternatives ignore strength and dosage form	Safety-checked substitution engine
5	Prescription scanning serves billing, not discovery	Prescription upload drives map search and whole-list matching

#	Gap in existing systems	MedNear feature
6	Failed searches are lost	Unmet-demand heatmap sent to nearby pharmacies
7	Limited data coverage	Owner-confirmed inventory that scales to any area

Gap statement: Existing systems either help patients find pharmacies or help pharmacies manage inventory, but none link OCR-driven stock updates, stock reliability scoring, prescription-aware safe substitution, and patient demand feedback into one closed-loop platform.

Solution and core modules

MedNear has three parts: a patient app, a pharmacy dashboard, and an AI layer that connects them.

1. Patient side

- Search by medicine name, or upload a prescription so OCR reads the medicines, strengths and quantities.
- Interactive map dashboard showing nearby pharmacies with availability, stock level, price, distance and directions.
- Price comparison across pharmacies, and a best-single-pharmacy match for a whole prescription.
- Reserve a medicine at a chosen pharmacy for a short time (see the hold feature below).
- Alerts when a searched medicine becomes available nearby.

2. Pharmacy owner side

- Inventory dashboard with manual add and edit.
- AI/OCR scanning of purchase bills and invoices, extracting medicine name, quantity, price, batch and expiry where available.
- Review-and-confirm step before any stock update is saved, to avoid OCR errors.
- Selling price and profit margin settings, kept within applicable price rules such as the MRP cap.
- Expiry and low-stock alerts, reservation inbox, and a demand view showing what nearby patients searched for.

3. AI layer

- OCR and document extraction for bills and prescriptions.

- Medicine name matching against a master drug database to correct OCR mistakes.
- Stock confidence scoring, substitution safety checks, and anonymous demand aggregation.

Novel features

Three features turn MedNear from a finder into a closed-loop system; stock confidence is the headline one.

1. Stock confidence score

OCR-based stock is a snapshot, so every result shows how reliable it is. The score uses data age, the medicine's sales velocity, and how regularly the pharmacy updates. Each result also shows a last-verified time, and the patient can send a one-tap confirm request to the pharmacy before travelling.

2. Safety-checked substitution engine

When the prescribed brand is missing, MedNear suggests substitutes by matching salt, strength and dosage form. It flags risky swaps such as extended-release against normal-release, shows the saving against the prescribed brand including Jan Aushadhi options, and always asks the patient to confirm with a doctor or pharmacist. It is a suggestion tool, not medical advice.

3. Unmet-demand heatmap

Every failed search is anonymised and aggregated by neighbourhood. Pharmacy owners see messages such as "15 people nearby searched for this medicine this week and you do not stock it", along with reorder suggestions. This uses the patient side to help the pharmacy side, which single-sided tools cannot do.

Reserve and hold

A held medicine is tracked as reserved stock, not removed from real stock, so physical counts stay correct and other patients see only what is truly free.

- **On hand:** physical units; changes only on a sale, a purchase, or a confirmed bill scan.
- **Reserved:** units held for patients.
- **Available to others** = on hand − reserved. This is the number the map shows.

Hold flow

1. The patient picks a pharmacy and quantity and requests a hold.
2. The system checks availability and locks the units in one database transaction, so two patients cannot claim the last strip at once.

3. The pharmacy is notified and confirms the item is set aside, which also verifies the stock.
4. The patient receives a pickup code and a deadline, for example 2 to 4 hours.
5. At the counter the pharmacist scans the code and marks the order collected; on hand and reserved both drop.
6. If the deadline passes or the patient cancels, the hold releases automatically and availability returns.

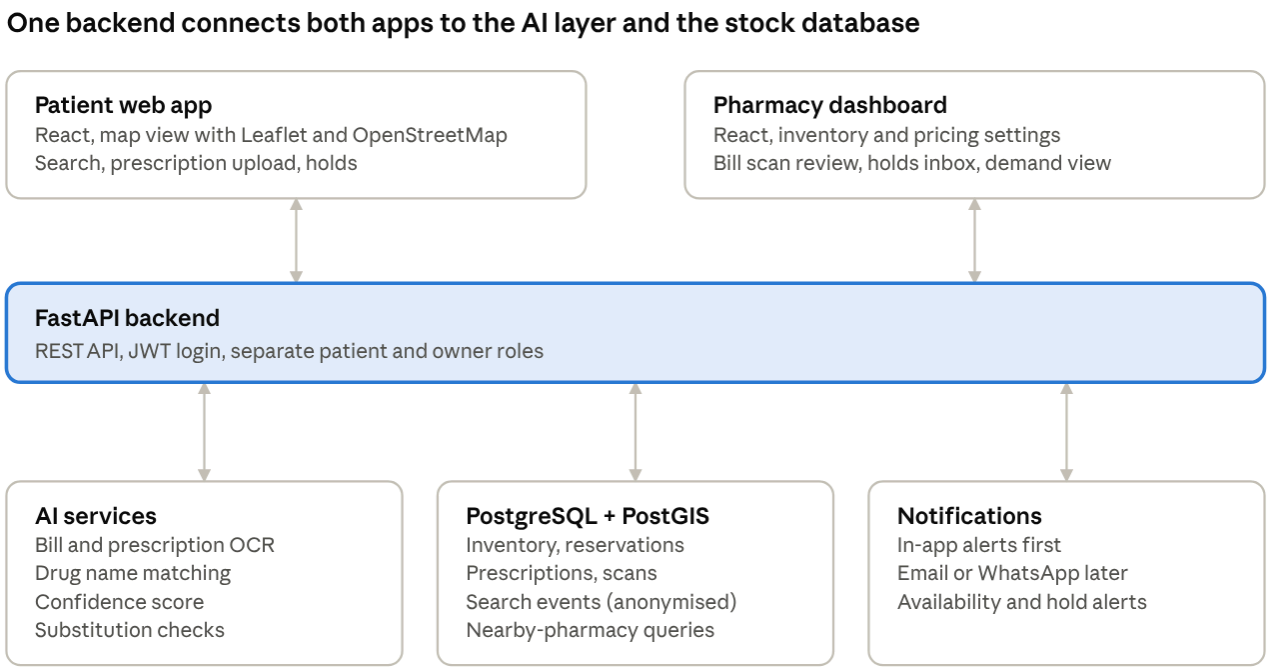
Rules

Question	Decision
Hold duration	Short and fixed, with one optional extension
Quantity cap	Limit per hold and per patient, to stop stock being locked out
Patient no-show	Tracked; repeat no-shows lose hold access
Pharmacy cannot fulfil	Hold cancelled, pharmacy reliability score reduced
Prescription-only medicines	Pharmacist still checks the prescription at pickup

Payment phasing: Phase 1 uses free holds with a timer and pickup code, to prove the stock logic. A paid token or full prepayment comes later through a licensed payment gateway, after checking the current rules on collecting payment for medicines.

Architecture and tech stack

Patients and owners use separate React front ends over one FastAPI backend, which calls the AI services, reads and writes the PostgreSQL/PostGIS database, and sends notifications.



system architecture · 2 apps, 1 backend, 3 services

This stack is a proposal; swap any layer if your course or team requires a different one.

Data model

Eight main tables hold the whole system; PostGIS handles the location queries.

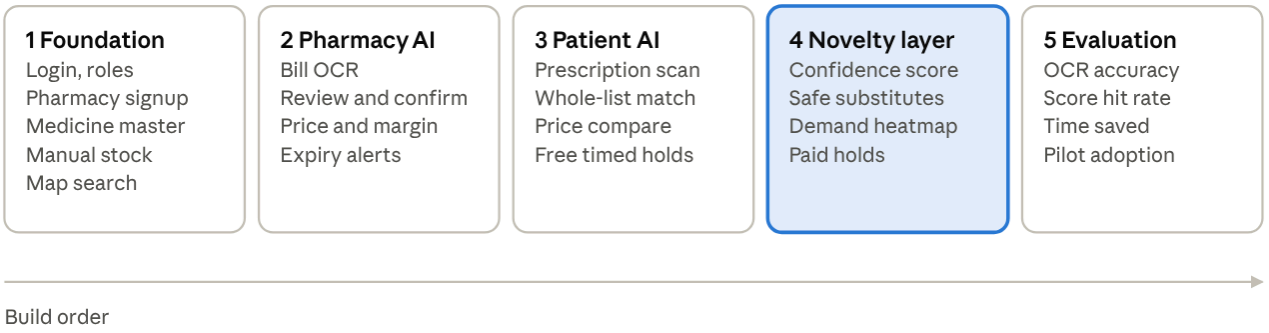
Table	Key fields	Purpose
users	id, role (patient or owner), name, contact	Accounts for both roles
pharmacies	id, owner_id, name, licence no., location (geo point), hours	Pharmacy profile and map position
medicines	id, brand, salt, strength, dosage form, MRP	Master drug database used for matching
inventory	pharmacy_id, medicine_id, on_hand, reserved, selling price, batch, expiry, last_verified	Stock per pharmacy; available = on_hand − reserved
scans	id, pharmacy_id, file, extracted items, status (pending, confirmed)	OCR bills and review state
prescriptions	id, user_id, file, extracted items, retention date	Uploaded prescriptions, short retention

Table	Key fields	Purpose
reservations	id, user_id, pharmacy_id, medicine_id, qty, status, expires_at, pickup code	Holds and their lifecycle
search_events	id, anonymised area, medicine_id, found (yes or no), timestamp	Source for the unmet-demand heatmap

Build roadmap

The build runs in five phases: core search and stock first, then the AI on each side, then the novelty layer, then evaluation.

Core search and scanning come first; the novelty layer builds on them



build roadmap · 5 phases

Paid holds wait until the free-hold flow has proven the stock logic.

Business model and pitch

Pharmacies are the paying customers and the data source, so the plan starts with a small pilot of local pharmacies before any wider launch.

Who to pitch

Audience	Why	Emphasise
Pharmacy owners	They supply the data	Less manual entry, expiry alerts, new walk-in customers
Incubators and accelerators	Funding, mentoring, credibility	Novelty, working prototype, pilot plan

Audience	Why	Emphasise
Healthtech angels and early VCs	Seed funding	Two-sided network effect, market size
Chemist associations and distributors	Can onboard many pharmacies at once	Digital visibility without extra work

Revenue options

- Pharmacy subscription for the inventory and OCR dashboard, with a free tier to attract owners.
- Small fee on paid reservations, added only after the paid-hold phase.
- Anonymised, aggregated demand insights for distributors.
- No paid ranking: results are ordered by availability and distance to protect patient trust.

Pitch structure: problem, solution, gap table, closed-loop flow, live demo, business model, market, competition, roadmap with pilot traction, and the ask. Look up current pharmacy-count and market figures from a reliable source before quoting them.

Risks, compliance and safety

The main risks are health-data privacy, unsafe substitution, and regulatory scope; each has a built-in control.

Risk	Control
Prescription data is sensitive health data	Encrypted storage, minimal retention, consent, aligned with India's DPDP Act; confirm requirements with a legal advisor
Wrong or unsafe substitute shown	Rules-based safety checks, "confirm with doctor or pharmacist" on every suggestion, never presented as medical advice
OCR errors corrupt stock	Mandatory review-and-confirm before any stock update; fuzzy matching to the master drug database
Stale stock misleads patients	Confidence score, last-verified time, confirm ping
Regulatory scope (e-pharmacy rules)	Position as discovery and inventory software, not a seller of medicines; check current regulations before launch

Risk	Control
Fake holds or hoarding	Hold caps, short timers, no-show tracking
Demand data exposes individuals	Anonymised, aggregated by area before any pharmacy sees it
Pharmacy price rules	Selling price limited to the MRP cap and applicable regulations

Evaluation metrics

Success is measured on the novel parts, not just on whether the app runs; targets are to be set after the first pilot.

Metric	What it shows
OCR field accuracy (medicine, quantity, price, expiry)	Whether bill scanning is reliable enough to trust
Time to update stock per bill, scan vs manual	Effort saved for pharmacies
Confidence-score hit rate vs real availability	Whether the headline novelty actually predicts stock
Prescription extraction accuracy	Quality of the patient-side OCR
Substitution suggestions accepted by pharmacists as safe	Safety of the substitution engine
Hold completion rate and no-show rate	Health of the reservation flow
Share of unmet searches later fulfilled after owner action	Value of the demand loop
Pilot pharmacies active weekly	Real adoption